IPv4 Addressing, Subnetting, CIDR & Fragmentation

Coverage: IPv4 20-60 Byte Header, Fragmentation & Reassembly, CIDR Slash Notation, Subnet Design, NAT & ICMP

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1. IPv4 Header & Fragmentation

- 20-60 Byte Header: IHL, Total Length, ID, Flags (DF/MF), Fragment Offset (8-byte units), TTL, Protocol, Checksum, IP addresses.
- Fragmentation occurs when datagram size > MTU.

2. CIDR & Subnetting

- CIDR prefix /N. Usable hosts = 2^(32 - N) - 2.

■ Worked Problem: IPv4 Fragmentation

- Given: 4000 byte packet (20B header + 3980B payload) on MTU = 1500 bytes.
- Frag 1: Len=1500, Payload=1480, MF=1, Offset=0 | Frag 2: Len=1500, Payload=1480, MF=1, Offset=185 | Frag 3: Len=1040, Payload=1020, MF=0, Offset=370.

■ KTU University High-Yield Exam Questions

- Q: Draw IPv4 header and solve a fragmentation numerical problem. (8 Marks)